Rethink the Noise Prior of Initialization Gap in Video Diffusion Models

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Abstract

Video Diffusion Models have advanced video generation by integrating text and image conditioning, offering enhanced control over generated content. However, maintaining consistency across frames remains a challenge, especially when using text prompts as control conditions. Some approaches, like FreeInit, address this by iteratively updating the initial noise to ensure video consistency, while methods like UniCtrl employ Attention Control to maintain spatiotemporal consistency. Yet, these techniques incur additional computational costs and inference time. Addressing the need for stable and consistent video generation without extra computational expense remains an open problem. In this paper, we revisit the noise prior to the initialization gap in video diffusion models and introduce a novel initialization method ***FastFreeInit****. By partially sharing the initial noise across different frames, we achieve enhanced consistency and stability in video generation without additional computational demands, as verified by our experiments.*

1. Introduction

Diffusion Models (DMs) have demonstrated superior performance in image synthesis, surpassing traditional methods such as GANs [12, 25, 26] and VAEs [27, 37, 49] in terms of stability and quality. Early research [19, 24, 29, 43–45] laid the essential groundwork for DMs, proving their effectiveness in scaling with varied datasets. Recent innovations [30, 32, 36, 38, 40, 58, 60] have enhanced their controllability and interaction with users, facilitating the generation of images that more accurately align with user specifications.

Recently developed Video Diffusion Models (VDMs) [20] have employed Diffusion Models (DMs) for generating videos. VDMs demonstrate their ability to produce videos depicting a range of motions in text-to-video synthesis tasks, facilitated by the integration of text encoders [35], as evidenced in works like [1, 2, 16, 18, 22, 59]. Various open-source text-to-video models have emerged, such as ModelScope [51], AnimateDiff [16], and VideoCrafter [6]. These models often rely on a pre-trained image generation model like Stable Diffusion (SD) [39] and incorporate additional temporal or motion components. Despite this, texts, unlike images which are rich in semantic content, struggle to maintain frame-to-frame consistency in video outputs. Concurrently, some research utilizes image conditions to foster image-to-video transformations with enhanced spatial semantic consistency [1, 15, 23]. While there are approaches proposing a text-to-image-to-video framework [11], relying solely on image conditions often falls short in controlling video motion. Combining both text and image conditions enhances spatiotemporal consistency in a mixed text-and-image-to-video process [6, 7, 14, 55, 61], though these techniques necessitate additional training.

Currently, Video Diffusion Models (VDMs) typically incorporate additional temporal layers into a 2D UNet; however, this modification fails to adequately address cross-frame constraints during the training of the 2D UNet model. Various training-free methods, such as those documented in recent research [8, 13, 34, 54], have attempted to improve the smoothness of generated videos by refining the start noise or taking the use of attention control [4, 17, 48, 57]. Despite these efforts, the challenge of maintaining consistent cross-frame coherence in videos produced by VDMs remains unresolved. In this paper, we reevaluate the noise prior to the initialization gap in video diffusion models and introduce a new initialization technique, **FastFreeInit**. By sharing initial noise partially across different frames, we achieve enhanced consistency and stability in video generation without imposing additional computational costs, as confirmed by our experimental results.

2. Background

Video Generation Numerous studies have explored the realm of video generation, employing various approaches like GAN-based frameworks [3, 42, 47] and transformer-based architectures [21, 50, 52, 53]. Building on the success of Diffusion models (DMs) [19, 24, 29, 43–45], which have delivered impressive outcomes in image synthesis [31, 32, 36, 38, 40], video diffusion models (VDMs) [20] have also showcased their prowess in generating videos

[2, 6, 7, 11, 14, 16, 18, 22, 41, 51, 55, 59].

Presently, VDMs typically integrate extra temporal layers into a 2D UNet, yet this adaptation does not sufficiently address cross-frame constraints during the training of the 2D UNet model. Several techniques [8, 13, 34, 54] have experimented with training-free approaches to enhance the smoothness of the generated videos. Nonetheless, the challenge of maintaining consistent cross-frame coherence in videos produced by VDMs persists.

Noise in Diffusion Models Only a few studies have highlighted the drawbacks in the noise schedules of existing diffusion models. In the realm of image synthesis, [28] identifies that traditional diffusion noise schedules do not entirely obscure the information in natural images, which constrains the model to produce only images of moderate brightness. Building on this, [9] further investigates the issue of signal leakage and introduces a method to explicitly model this leakage for an improved inference noise distribution, resulting in images of greater brightness and color diversity.

In the context of video, PYoCo [10] meticulously formulates a progressive video noise prior to enhanced video generation. Echoing [28], PYoCo also emphasizes noise schedule adjustments during the training phase and necessitates extensive fine-tuning on video datasets. Recent initiatives [13,34] similarly focus on the initial noise at inference, albeit with a goal of producing longer videos. FreeInit [54] aims to elevate inference quality and further incorporates tailored frequency-domain operations to adjust various frequency components of the initial noise, but it needs additional computational costs and inference time.

3. Method

Although Video Diffusion Models (VDMs) have achieved notable success in video generation, most open-source VDMs still struggle with consistency and stability in their generated videos. Research has indicated that the consistency of VDM-generated videos can be influenced by the initial noise [54], while inconsistencies in the attention mechanism's value can lead to unstable video outputs [8]. FreeInit [54] addresses this by iteratively updating the initial noise to ensure consistency across frames, whereas UniCtrl [8] enhances video quality by managing the consistency of values within the attention mechanism. However, these methods require additional inference time and computational resources. FreeInit involves multiple sampling processes, and UniCtrl necessitates concurrent inference across multiple branches. Given the substantial memory and computation demands of VDMs, these additional costs are often impractical. Therefore, exploring ways to enhance the consistency and stability of video generation without increasing computational costs remains a pressing issue. Our approach, inspired by PYoCo [10], mixes the noise from the first frame with that of subsequent frames and, following FreeInit [54], blends this combined noise at various frequencies. This method achieves improved spatio-temporal consistency in generated videos without additional computational overhead.

Noise Mixing Consider the scenario where $\varepsilon^1, \varepsilon^2, \ldots, \varepsilon^{n_s}$ represent the specific noises associated with each frame of a video, with ε^i being the ith noise element in the noise tensor ε. Following PYoCo [10], we define two distinct types of noise vectors: $\varepsilon_{\text{shared}}$ and ε_{ind}. The vector $\varepsilon_{\text{shared}}$ serves as a universal noise component common to all frames, whereas ε_{ind} consists of unique noise vectors tailored to each individual frame. These two vectors are then combined linearly to form the noise applied to each frame.

Mathematically, this is described by the following formulation:

$$\varepsilon_{\text{shared}} \sim N\left(0, \frac{\alpha^2}{1+\alpha^2} I\right), \varepsilon^i_{\text{ind}} \sim N\left(0, \frac{1}{1+\alpha^2} I\right) \quad (1)$$
$$\varepsilon^i_{new} = \varepsilon_{\text{shared}} + \varepsilon^i_{\text{ind}}$$

This structure ensures that while each frame benefits from a base level of commonality due to the shared noise, it also preserves a degree of uniqueness through the individual noise components, thus balancing consistency and variation across the video sequence. Here, we use the noise of the first frame from the latent as our shared ε.

Noise Reinitialization Like FreeInit [54], we utilize a spatio-temporal frequency filtering technique, whereby we amalgamate the low-frequency elements of the original noise latent vector z_T with the high-frequency elements of a newly generated random Gaussian noise η. This yields a dynamically reinitialized noisy latent vector z'_T. By employing this method, we retain the critical information embedded within the low-frequency components of z_T, while infusing variability in the high-frequency spectrum to enrich visual textures and details. The formulation of this process is detailed in the following mathematical expressions:

$$\mathcal{F}^L_{z_T} = \mathcal{FFT}3D(z_T) \odot \mathcal{H}, \quad (2)$$
$$\mathcal{F}^H_{\eta} = \mathcal{FFT}3D(\eta) \odot (1 - \mathcal{H}), \quad (3)$$
$$z'_T = \mathcal{IFFT}3D(\mathcal{F}^L z_T + \mathcal{F}^H \eta), \quad (4)$$

Here, $\mathcal{FFT}3D$ denotes the Three-Dimensional Fast Fourier Transformation applied across both spatial and temporal dimensions, and $\mathcal{IFFT}3D$ is the Inverse Fast Fourier Transformation that reconstructs the combined latent z'_T from the frequency domain to the time-space domain. The filter $\mathcal{H}$ represents a Spatial-Temporal Low Pass Filter (LPF), designed to match the dimensions of the latent, facilitating selective frequency blending.

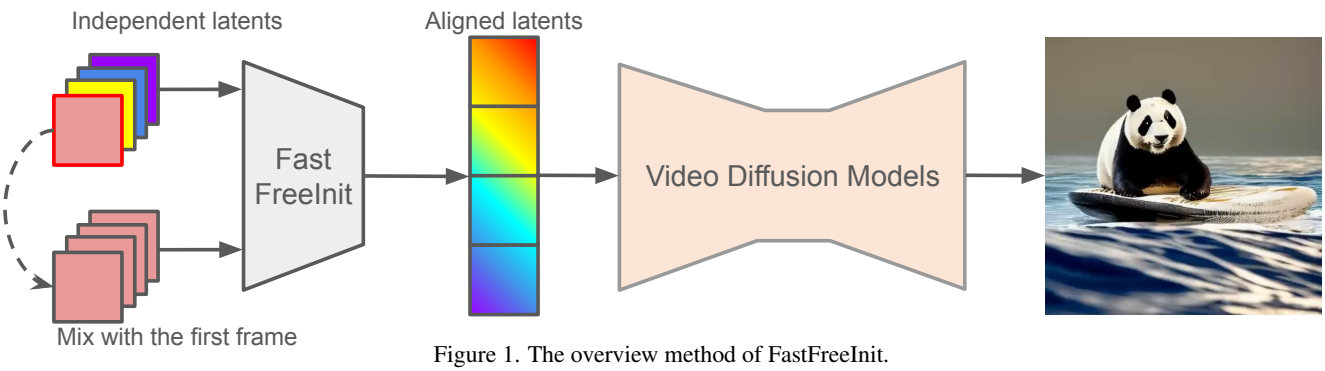


Figure 1. The overview method of FastFreeInit.

FastFreeInit To improve the consistency of videos in text-to-video tasks, we focus on manipulating the initial noise. According to Ge et al. [10], controlling the initial noise may sometimes result in sub-optimal consistency of the generated videos. Our method for manipulating initial noise is structured into two distinct steps:

1) Adopting the Mixed Noise Model methodology proposed by Ge et al. [10], we control the noise for each frame as comprising both shared and independent components. The shared component is directly sourced from the noise of the first frame, while the independent component consists of initial random noise. Consequently, the noise for each subsequent frame is composed of the first frame's noise combined with its original random noise. The corresponding formula is presented as 1 and the formula applied is

$$\varepsilon^i_{new} = \frac{\alpha^2}{1+\alpha^2} * \varepsilon^1 + (1 - \frac{\alpha^2}{1+\alpha^2}) * \varepsilon^i_{\text{orignal}}$$

2) Building on the findings of Wu et al., substantial spatio-temporal correlations exist within the low-frequency components, we extend those correlations from the first frame to others like 2. Therefore, we preserve the low-frequency components of the noise processed in the first step, adding the original latent high-frequency components.

Here is the full algorithm:

Algorithm 1 FastFreeInit Algorithm

1: $latents \leftarrow random\ noises for all frames$
2: $latents_{low} \leftarrow the\ first\ frame\ of\ latents$
3: $a \leftarrow \frac{\alpha^2}{1+\alpha^2}$ and $b \leftarrow 1 - a$
4: $latents_{low} \leftarrow a \times latents_{low} + b \times latents$
5: $latents \leftarrow freq_mix_3d(latents_{low}, latents)$

Implement Details We began with the official repository of FreeInit [54] and integrated our FastFreeInit pipeline based on the FreeInit template, incorporating an initial noise control algorithm prior to the denoising step. We utilized the formula and the freq_mix_3d function from FreeInit to manage the noise control. We ensured that the initial random noise was consistent across all three approaches and timed them within the main program. The core code is detailed in the appendix.

For the evaluation phase, we employed relevant models from Hugging Face to assess various metrics. We simultaneously generated and evaluated those videos with the prompt text. The relevant code is provided in the appendix for further reference.

4. Results

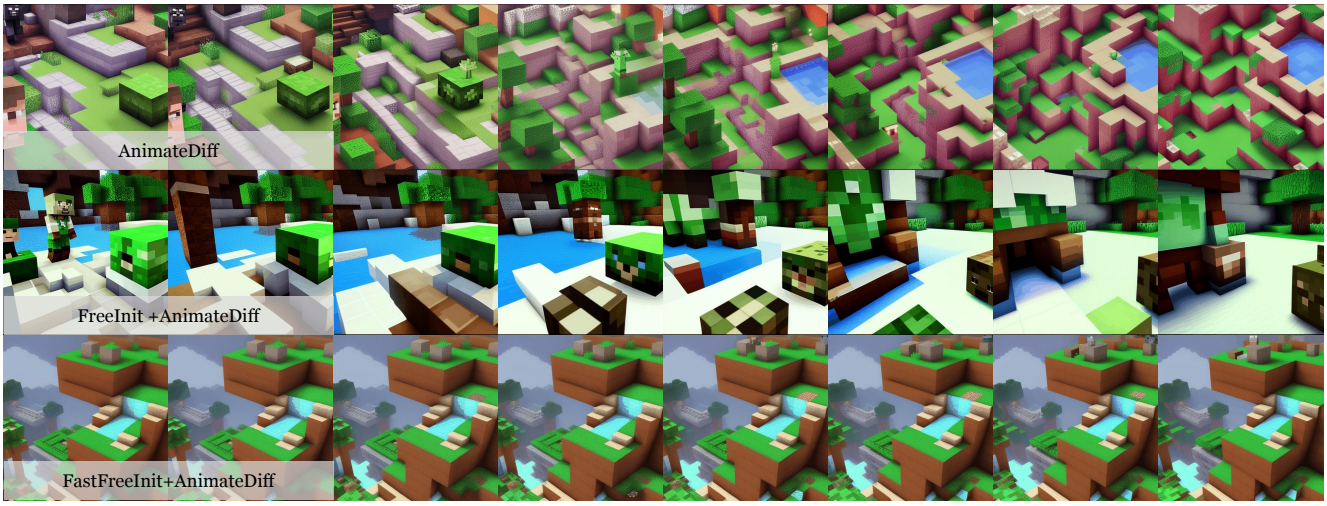
To assess the effectiveness of our model, we utilize prompts from two datasets: UCF-101 [46] and MSR-VTT [56] to generate videos. In line with the approach of Ge et al [10] and Chen et al [8], we employ identical prompts from the UCF-101 dataset for our experiments. Additionally, we select 100 unique prompts from the MSR-VTT dataset to further evaluate our model. These selections form our comprehensive dataset for evaluation. Next, we provide a brief introduction to evaluation metrics and backbone.

4.1. Metric

To quantitatively measure the performance of our model, we employ standard metrics as outlined in [41, 54].

• *Clip Similarity*: To measure the relevance between videos and texts, we compute the average Clip Similarity [52] for each generated video with their corresponding prompt. We calculate the score by averaging Clip Similarity for each frame in the video with its prompt. In our experiment, we compute the CLIP similarity utilizing TorchMetrics with clip-vit-base-patch32 model [35].

• *DINO*: To assess the spatiotemporal consistency of the generated videos, we utilize DINO [33] to compute the cosine similarity between the initial frame and subsequent frames. The average DINO score across all consecutive frames is then used as the video's overall score. In our experiments, we utilize the DINO-vits16 [5] model to compute the DINO cosine similarity.



A man narrates his minecraft gameplay

A car is shown

The woman sings into the microphone

Figure 2. Qualitative Comparisons

Table 1. Quantitative Comparisons on UCF-101 and MSR-VTT. FastFreeInit significantly improves the temporal consistency without adding too much extra time for generating video. I indicates the number of iterations for FreeInit.

Method	CLIP (↑)	DINO (↑)	Time (↓)
AnimateDiff [16]	95.18	97.18	74.40s
FreeInit + AnimateDiff ($I = 3$)	96.95	98.32	218.91s
FastFreeInit + AnimateDiff	$97.69_{(+00.74)}$	$99.11_{(+00.79)}$	$74.40s_{(-141.51s)}$

4.2. Backbones

Given the plug-and-play nature of our approach, we opted to test our methods using AnimateDiff. *AnimateDiff* [16] provides a practical framework to impart motion dynamics into personalized text-to-image models like those developed through Stable Diffusion. This is achieved without necessitating adjustments specific to each model. At the core of AnimateDiff lies a motion module. Once trained, this module can be universally applied to various personalized text-to-image models that share the same foundational model, utilizing transferable motion priors derived from real-world videos to enable animation.

4.3. Baseline

We chose FreeInit as our baseline because it is a training-free method that aims to enhance the appearance and temporal consistency of generated videos. It does this by iteratively refining the spatial-temporal low-frequency components of the initial latent code during inference. Given that both FreeInit and FastFreeInit are training-free methods designed to improve spatiotemporal consistency in video generation through diffusion models, it is logical to compare the performance of FastFreeInit against FreeInit.

4.4. Qualitative Comparisons

For qualitative comparisons, shown in Figure 2, reveal that our FastFreeInit method markedly improves spatiotemporal consistency and adheres closely to the text. For instance, using the text prompt 'A man narrates his Minecraft gameplay', the standard AnimateDiff method would cause abrupt transitions from a green-black box to a blue pool. In contrast, FastFreeInit maintains greater consistency with the elements. Additionally, FreeInit would have the ground details change inconsistently, whereas FastFreeInit preserves better consistency in these feature details as well. Furthermore, FastFreeInit is more aligned with the prompt, as it avoids visualizing the narrator, which is implied by the text that the man should not appear in the video. Both Normal AnimateDiff and FastFreeInit cause Minecraft characters to appear in the video.

4.5. Quantitative Comparisons

For quantitative comparisons, the results for UCF-101 and MSRVTT are presented in Table 1. We compare the base backbone with it augmented by FastFreeInit and FreeInit, respectively. The result demonstrates that our FastFreeInit method significantly enhances spatiotemporal consistency and preserves the meaning of text within the video. Additionally, it offers substantial time savings compared to the FreeInit method while improving both CLIP and DINO scores. The improvement in the CLIP score from 96.95 to 97.69 suggests that our method adheres more closely to the prompt instructions than the original method and FreeInit. Furthermore, the enhancement in the DINO score from 98.32 to 99.11 indicates that our method improves video consistency over the other two methods. In conclusion, our FastFreeInit method outperforms the other approaches in these three critical aspects.

5. Conclusion

We present FastFreeInit as a novel solution aimed at enhancing cross-frame consistency and stability in Video Diffusion Models without the need for additional training. By ingeniously managing the initial noise distribution across frames, FastFreeInit significantly improves the spatiotemporal consistency in generated videos. This method is distinguished by its ease of integration with existing models and does not require extensive fine-tuning, ensuring broad applicability across different video generation frameworks. The performance of FastFreeInit has been thoroughly validated through rigorous testing, confirming its effectiveness and showcasing its potential as a versatile tool for video generation models.

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A. Core Algorithm Code

```
latents_low = latents[:, :, 0, :, :]
z_T = latents_low.unsqueeze(2).expand(-1, -1, latents.shape[2], -1, -1)

alpha_sqr = alpha * alpha
a = alpha_sqr / (1 + alpha_sqr)
b = 1 - a
z_T = a * z_T + b * latents

latents = freq_mix_3d(
    z_T.to(dtype=torch.float32), latents, LPF=self.freq_filter
)
```

B. Evaluation Code

```
device = torch.device('cuda' if torch.cuda.is_available() else "cpu")
processor_clip = CLIPProcessor.from_pretrained("openai/clip-vit-base-patch32")
model_clip = CLIPModel.from_pretrained("openai/clip-vit-base-patch32").to(device)

processor_dino = ViTImageProcessor.from_pretrained("facebook/dino-vits16")
model_dino = ViTModel.from_pretrained("facebook/dino-vits16").to(device)

def clip_score(outputs, processor, model, device):
    image_features = []
    with torch.no_grad():
        for output in outputs:
            input_tmp = processor(images=output, return_tensors="pt").to(device)
            image_feature = model.get_image_features(**input_tmp)
            image_features.append(image_feature)
        cos = nn.CosineSimilarity(dim=0)
        Clip = 0
        for i in range(1,len(image_features)):
            sim = cos(image_features[i-1][0],image_features[i][0]).item()
            sim = (sim+1)/2
            Clip += sim
        return Clip / 15

def dino_score(outputs, processor, model, device):
    image_features = []
    with torch.no_grad():
        for output in outputs:
            input_tmp = processor(images=output, return_tensors="pt").to(device)
            image_feature = model(**input_tmp).last_hidden_state
            image_feature = image_feature.mean(dim=1)
            image_features.append(image_feature)
        cos = nn.CosineSimilarity(dim=0)
        Dino = 0
        for i in range(1,len(image_features)):
            sim = cos(image_features[i-1][0],image_features[i][0]).item()
            sim = (sim+1)/2
            Dino += sim
        return Dino / 15
```